ENGR-095
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Finishing our successful prototype for ENGR-095, and completing our project to all of the specifications, requirements and constraints to create our bookkeeping and preservation system was an incredibly successful, and satisfying conclusion to the first half of the module semester. Initially, when I started this project, I was unsure about the nature of our device, and even what to make. Even when we figured out our design and conceptual idea, as well as beginning our planning for how to execute our ideas and create our vision, I was unsure if our idea was "impactful" enough- or even mattered enough in the grand scheme of things compared to other individuals projects and work.

When beginning the engineering process of our project, we ran into a lot of initial roadblocks- most notably, our sensors did not work how we wanted them to, or even assumed they would. Our iterations went through several stages- figuring out that the light sensor wouldn't sense light- and trying to perhaps correct it to sense darkness. This however, was shot down- as we realized that the sensor could in fact sense light- the tolerance / calibration of the device simply had to be adjusted. The more we became more familiar with our sensors and the technology / equipment at our disposal, I found myself growing more and more comfortable with constructing, and using the things that were given to us.

There were however, some sensor related setbacks- as well as pieces being moved around, swapped out and reused. Initially, our group was using the electronic moisture sensors- the ones supplied to us during the earlier experiments we had wired up. However, we soon found them to be quite faulty as many other groups did. We began getting copious amounts of false results and positives, and no consistent baselines for our sensor. So, as a result, we had to make the switch to our handmade sensors- which presented new challenges with that calibration. The way our code functioned, the servo would move when the light or moisture reached a certain value / threshold. However, our sensors kept giving us varying results- slightly different, every single time. So, we had to get creative. The more we got used to our electronics, the better we were able to predict their values, and eventually found concrete thresholds that would indicate the servo to move, and were consistently successful, despite how non-ideal the circumstances of our technology was.

As we moved through the project, we found that most of our issues were generally within the minute coding details. As far as our physical construction and wiring, we were able to do those quite easily, with our hands-on knowledge, and some assistance from AI for the most complex setups of wires when we were not explicitly aware of which labels meant what. As we moved forwards, we found that we consistently began having issues with our LED- moreover, we couldn't really get it working. It would of course turn on when connected to the 5.5V- so it wasn't shorted. It just wasn't reacting to the signal. I came during office hours, believing it to be a larger code issue- however, upon receiving assistance, found it was human error. The labels on the arduino were different from the PIN number in the code. Throughout the module, I had been having specific issues with turning on LED's with code- as they just weren't connected to the new PIN- and I had no idea why. Well, put quite simply- now I know.

Throughout this project, as well as this module, I learned a lot about Engineering- and also the college experience. I cannot put into words just how the shift from college to high school has been. It has been difficult, and I've found it hard to organize myself. This has resulted in me missing deadlines- I didn't have a schedule for things. I wasn't organized quite in the way I wanted to be. But, as a result, I have to make changes. Just like engineering, I have to adapt. I cannot use my early assumptions, and I have to keep moving even when my designs fail. Like the LED PIN issue, sometimes it's a simple snap of the fingers. Other times, like with the electric to handheld sensor switch, it's a bit of a tougher, larger problem. Through this module, it showed me I have to keep going- and put in more work than I think. I have to be present, and create!

The first half of this module and the final project have been fantastic- I have found a love for hands-on engineering- circuitry work and manufacturing our physical projects. Through this FYRE project, I have made new friends, a pretty neat mechanism for storing books, and gained vast engineering knowledge and gotten closer to what I love about engineering.

Ultimately, this project has been incredibly enlightening, and helped me greatly. While our project was successful in all of the constraints and requirements, it was successful in ways more than that- as it taught me about the concepts and values of engineering.